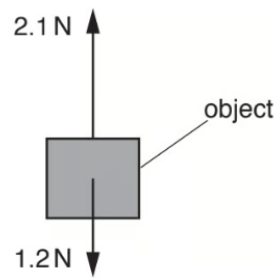


3. The diagram shows two opposite vertical forces of magnitude 1.2 N and 2.1 N acting on an object.



Which of the following statements is wrong?

- A** The magnitude of the resultant force is 0.9 N.
- B** The object is accelerating and moving up.
- C** The object is decelerating and moving up.
- D** The object is decelerating and moving down.

Ans: C

Option A: The magnitude of the resultant force is $2.1 - 1.2 = 0.9$ N

Option B: The resultant force is 0.9 N upwards since the force exerted on it upwards is larger than the force exerted on it downwards. If object is moving upwards in the same direction of the resultant force, it will be accelerating.

Option C: If object is moving upwards in the same direction of the resultant force, it will be accelerating and not decelerating. So this is the wrong statement.

Option D: If object is moving downwards in the opposite direction of the resultant force, it will be decelerating.

4

4 The diagrams show a hollow metal cube with a sealed opening is suspended from a